Report of Findings

The dataset was analyzed using various statistical and visualization techniques to understand patterns and relationships.
Missing values in the 'age' column were filled using median imputation.
Key insights obtained from the analysis are:

1. Most passengers were between the age group of 20-40. The distribution is slightly right-skewed, indicating fewer older passengers.
2. More passengers did not survive compared to those who survived.
3. Females had a significantly higher survival rate compared to males.
4. Passengers in first class had higher survival rates compared to second and third class passengers.
5. Younger passengers had a slightly higher chance of survival compared to older passengers.
6. Survival has a positive correlation with fare and a negative correlation with passenger clas
7. Pairplot shows relationships between variables, where survival is more common among passengers with higher fare and lower class number.
8. Strong relationships were observed between passenger class, fare, and survival through correlation analysis.
9. First-class passengers had the highest survival rate, confirming socioeconomic influence on survival.

Overall, the analysis highlights that gender, class, and fare were the most influential factors affecting survival.This analysis can help in building predictive models for survival classification.